



Empirical analysis of economic performance and metal prices in Peru using svar and okun's law

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Abstract

Observing the growth and improvements in the Peruvian labor market over the past 27 years, this study aims to test the validity of Okun's Law hypothesis for Peru during the period 1992–2019. To achieve this, a novel methodology is proposed, which involves calculating the production gap and unemployment gap. Firstly, a proxy variable for the time-varying natural rate of unemployment and a measure of potential output are calculated, both following the methodological approach suggested by the literature. Subsequently, a structural vector autoregressive model (S-VAR) is estimated to capture the lagged effects of Okun's theoretical relationship. The main findings of the study suggest that shocks in the precious metal price index impact both the production gap and the unemployment gap, confirming that the international dynamics of precious metal prices play a crucial role in economic performance when analyzed from the perspective of Okun's Law. This article demonstrates, in the context of a Latin American country, that Okun's law can be fulfilled by considering a fundamental analysis of the determinants of economic growth.

Keywords SVAR · Product gap · Unemployment gap · Commodities, Okun's Law

Introduction

The Peruvian economy achieved its economic stability program in 1991, which was important for reducing inflation. After three years of repeated GDP declines, a moderate stability was established with a slight growth. In 1992, the productive activity was severely affected by extraneous factors that restricted the aggregate supply, but an adjustment

program led to the lowest inflation rate in the past 15 years. From 1993 to 1998, the substantial reduction of inflation and the introduction of consistent fiscal and monetary measures allowed for the lowest inflation rate since 1977. The period from 1999 to 2001 allowed for a 3.1% increase in GDP on average, with a trend exceeding 5%, due to external policies and investor expectations. Overall, the decade was characterized by an environment of stability, accompanied by measures to control prices and economic stabilization (Central Reserve Bank of Peru 2021).

Similarly, the period from 2002 to 2013 showed remarkable economic dynamics, with sustained growth of real capital stock above 6% and a sustained decrease in the unemployment rate. This was according to data from CEPAL and data analyzed from the Penn World table at the University of Groningen. The dynamic economic growth can be attributed to the adoption of macroeconomic stabilization policies aimed at stimulating medium-term industrial productivity levels in the nation through the achievement of external balance, controlled inflation, and fiscal discipline for controlling public debt. However, for the period from 2014 to 2017, a contraction of approximately 3.2% in economic growth was observed, mainly due to the international reduction in commodity prices, which affected the national income

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generated by the export of copper and other mineral exports of the Peruvian economy.

For the period from 2017 to 2019, after the 2.5% economic decline in 2017, the outlook for the economy in 2018 was hopeful with a 4% increase in activity driven by domestic demand and price regulation. In 2019, the economic activity grew by 2.2%, leaving behind the effects of the previous expansion due to the reduction of public spending in the country (Central Reserve Bank of Peru 2021).

However, the period of expansion of the Peruvian economy seems to have generated important medium-term results in the dynamics of economic growth, as highlighted by Mendoza Bellido (2017, p. 107). During the period from 2003–2012, there was a notable increase in the price of traditional exports, mostly mining of precious metals, averaging 15 percent annually, falling to 7 percent per year between 2013 and 2016 (due to the decline in international commodity prices). However, during the boom, private investment grew at an annual rate of 14 percent, and GDP grew at an average rate of 6.3 percent per year. The growth in private investment was accompanied by a contraction in unemployment, suggesting the hypothesis that an increase in economic performance is related to the dynamics of the mining sector and international prices of precious metals, ultimately impacting unemployment behavior."

In general, the literature has established a long-term relationship between the dynamics of economic growth and labor market outcomes, especially unemployment. The objective of various studies addressing this relationship has been to determine the GDP growth rate necessary to reduce the unemployment rate by a certain percentage. This relationship was originally investigated by Okun (1962), and subsequently addressed by various empirical studies that seek to confirm or refute the Okun hypothesis in different international contexts. As stated by Rodríguez and Peredo (2007, p.60), the hypothesis formulated by Okun is based on data from the 1950s for the US economy, and warns that this only constitutes an observation of an empirical regularity (of the real world). However, this regularity has stood the test of time.

In studies associated with estimating the output-unemployment relationship in the case of Peru, the work of Garavito (2003) stands out. The study manages to determine that there is a positive long-term relationship between the employment rate and output, which can be contradictory and may be due to changes in labor legislation, a fact highlighted by the author. He indicates that the low level of the Okun coefficient may be explained by the low sensitivity of the unemployment rate to changes in output, even in the presence of a contradictory relationship. In general, in the few studies on the Okun's Law estimated in the case of Peru, it has been found that the incidence of output on unemployment is not significant and the relationship may even not

hold. On the other hand, using a gap model, Vasquez (2018) identifies an inelastic or insensitive behavior. However, the methods used so far do not analyze the lagged and dynamic effect in the framework of dynamic multi equation models that may be more useful in capturing the Okun relationship under an accumulated lagged adjustment perspective. Thus, under a structural vector autoregressive (SVAR) methodology, the following research question arises: What is the impact of investment and the output gap on unemployment variation in Peru for the period 1992–2017? To answer this question, the present study is based on the relationship described by Okun (1962) as the theoretical foundation and presents a current estimation of Okun's Law for Peru, capturing the lagged effects of the endogenous variable system and considering a theoretical framework based on the previous proposal. The present study is innovative since it proposes a transitive functional form between the Metal price index, the output gap, and the unemployment gap. Thus, this is the first empirical study that proves said relationship in Peru under the methodology proposed.

The study also uses a methodological innovation based on calculating the output gap to specify a dynamic multi-equational system, based on the SVAR methodology, for the period 1991–2019. In accordance with the proposed objectives, this research article is structured as follows: Section (ii) presents the literature review, Section (iii) describes the research methodology in detail, Section (iv) presents the results and discussion, Section (v) presents the general conclusions of the study, as well as the final appendix of relevant estimates at the end of this work.

Literature review

Okun's law

Understanding the dynamics of unemployment behavior is a central objective for policymakers since this variable maintains a close relationship with a nation's resources and productive capacity. This is why the characteristics of unemployment by regions and countries are quite dissimilar (Blanchard and Galí, 2008). They emphasize the need to analyze the rigidities and frictions of the labor market and their cyclical effects to understand the behavior of unemployment in the short and medium term. However, it is undeniable that the behavior of unemployment depends on the rational decisions made by agents under criteria of wages and vacancies (Neffa et al. 2007), but it will also depend fundamentally on the macroeconomic cyclical fluctuations that affect the long-term states of the unemployment rate.

It should be noted that in the 1960s, a theoretical proposal emerged that aimed at the purpose described above.

The proposal sought to relate the level of cyclical economic activity in the goods and services market with the level of activity in the labor market, giving rise to "Okun's Law," an empirical macroeconomic relationship between production and unemployment. Okun (1962) presents his article in which he studies the inverse linear relationship between the variation of production and unemployment for the US economy, using corresponding data during the 1950s and highlighting two econometric specifications of interest. The first of these is a Difference Model:

$$u_t - u_{t-1} = \beta_0 + \beta_1 (y_t - y_{t-1}) + \varepsilon_t \quad (1)$$

And the second one, a Gap Model (Trial Gaps):

$$[u_t - u_t^* = \beta_0 + \beta_1 (y_t - y_t^*) + \varepsilon_t] \quad (2)$$

These empirical estimates of the Okun coefficient are based on recognizing the dependence between GDP (y_t) and the unemployment rate (u_t). The coefficients are interpreted as a measure of the responsiveness of unemployment to changes in GDP and vice versa, and are important in indicating the cost of unemployment in terms of aggregate output. As a result, various economists have sought to test Okun's postulates (1962) through empirical estimates, with seminal works including contributions from Smith (1975), Gordon (1984), and Knoester (1986). Specifically, Gordon (1984) decomposes real national output growth into a cyclical component and a trend or "potential" component, which is consistent with Evans' (1989) intention to estimate a VAR model to understand the effects of unemployment shocks on production. He finds that there is a feedback effect between GDP growth and unemployment, demonstrating a high degree of persistence in the long-term relationship between GDP growth and unemployment. These conclusions, in turn, form the scientific basis for the subsequent works of Prachowny (1993), Weber (1995), and Moosa (1997), whose results encouraged the growth of methodological applications based on the relationship predicted by Okun (1962).

Thus, a widely debated discussion in macroeconomic literature still raises questions about its empirical usefulness, as exemplified by Attfield and Silverstone's (1998) argument. In this work, the authors question whether production and unemployment are indeed co-integrated in the long term, and if so, the Okun coefficient can be interpreted as the long-term cointegration coefficient between the variables, serving as evidence of a long-term relationship.

Therefore, subsequent empirical works have mostly incorporated co-integration tests, which in some cases provide empirical evidence of the long-term relationship described by Okun. However, estimates of the Okun coefficient seem to vary substantially from country to country, or as expected, from region to region. To expand this view of the variability of results, readers are encouraged to review the works of Lee

(2000), Harris and Silverstone (2001), Sögner and Stiassny (2002), and Villaverde and Maza (2009). On the other hand, Knoester (1986) criticizes the original form of the Law proposed by Okun in 1962, noting that Okun's seminal results lack explicit treatments associated with classical causes of unemployment. In this sense, Freeman's (2001) study is one of the most important. In this research, the author delves into trend/cycle decomposition analysis to test Okun's Law in the context of a panel of 10 industrialized countries. In this analysis, he estimates Okun's gap model (1962) using the bandpass filter to investigate in greater depth the long-term cyclical determinants of unemployment (or classical causes) and their relationship to production.

Therefore, the use of the filter method in time series techniques to test short and long-term relationships between output and unemployment has become popular. Notable analyses have been conducted by Cuaresma (2003) and Apergis and Rezitis (2003), who estimate the Hodrick-Prescott decomposition filter to estimate the Okun's Law for the North American context. In this spatial context, Holmes and Silverstone (2006) highlight that there may be a significant inverse relationship between cyclical output change and unemployment during periods of economic expansion in the United States, for example, during the post-war period. For this reason, in a different context, Gordon (2010) suggests that it is necessary to quantify the response of cyclical changes in output to changes in employment as an alternative to traditional static analysis. Therefore, he proposes incorporating labor force participation and productivity as control variables of the business cycle. The author finds that the higher presence of white-collar workers and the lower dependence on unskilled labor explain why the Okun's Law lacks predictive ability for certain periods of time, a critique that is also recognized in the work of Owyang and Sekhposyan (2012). Likewise, in the context of the United States, Dell'Anno and Solomon (2008) estimate a structural equation to evaluate whether there is evidence of the fulfillment of Okun's Law, considering the GDP growth rate and the unemployment rate. The main finding of the authors provides evidence in favor of the hypothesis described by Okun.

Aligned with the methodological approach of this study, Karfakis, Katrakilidis, and Tsanana (2014) examine the relationship between unemployment and output for the Greek economy, using a VAR methodology. They question whether changes in the unemployment rate are systematically related to changes in output, finding through impulse-response functions that a persistent negative shock in demand (characteristic of a recession of real output) impacts unemployment, leading it to persistently high levels. This is evidence of an asymmetric unemployment-output relationship.

In this sense, Ball, Leigh, and Loungani (2017) effectively highlight asymmetric results regarding the fulfillment of Okun's Law for 20 developed economies, emphasizing

the marked differences between countries due to their economic structure and labor market composition. Nevertheless, the fulfillment of Okun's Law is evidenced. On the other hand, in the Latin American context, the work carried out by Molero (2012) studies the fulfillment of Okun's Law in first differences for the Venezuelan economy during the period 1999–2009, proposing various functional forms for linear models. They find that the dynamics of unemployment in Venezuela are conditioned by factors exogenous to output fluctuations.

Under an analysis of labor flexibility in Mexico, Loria, Ramirez, and Salas (2015) estimate an Error Correction Vector Model (VECM) that includes a labor flexibility index. The main result of the article indicates that an increase in labor flexibility in the formal labor market tends to increase the unemployment rate as a result of a high positive elasticity on unemployment that can have an important impact on long-term output growth. Finally, Sanchez (2015) questions whether economic growth induces a decrease in the unemployment rate in the Dominican Republic and attempts to empirically respond through an estimation of Okun's Law, proving that economic growth does not explain the change in the unemployment rate empirically and that this phenomenon can be attributed to other cyclical aspects of the Dominican economy.

On the other hand, Villavicencio and Ochoa (2017) test Okun's hypothesis for 12 Latin American countries using a Vector Autoregression (VAR) model of Panel data. Their main findings show that there is an inverse relationship between unemployment and economic growth in Latin American countries, and they also find evidence of a lagged effect that justifies implementing a functional form that captures lagged effects on the set of variables as proposed by VAR models. This approach will be applied in this study following a perspective of structural analysis of shocks.

Commodity prices in the context Latin America

In the mid-twentieth century, German economist Hans Singer (1950) studied the evolution of commodity prices during the period leading up to World War II and observed a downward trend in their terms of trade. Given the great importance of commodities in Latin America's exports, this empirical finding caught the attention of some researchers in the region. Raúl Prebisch, then executive secretary of the newly created Economic Commission for Latin America (ECLA), decided to continue studying the price relationship between commodities and manufactured goods. Prebisch's research results (1950) proved to be consistent with Singer's findings, and thus the hypothesis of the secular deterioration of commodity prices, also known as the Prebisch-Singer thesis, emerged.

Currently, although it is known that commodities do not necessarily exhibit a secular deterioration in their terms of trade year after year, the debate about the existence of a negative and statistically significant trend in the long term is still relevant and generates academic interest (see Harvey et al. 2010). It is also pertinent to note that today the number of methodological approaches aimed at studying the Prebisch-Singer thesis is broader than 70 years ago. In this line, Erten and Ocampo (2012) study the average duration of commodity super-cycles other than oil and show a decreasing behavior in the temporal interval that observes these persistent increases throughout the period 1865–2009. The authors conclude that although commodity price super-cycles still occur in global markets, they are becoming weaker.

However, in the twenty-first century, the slope of the price of some relevant commodities is clearly positive. This situation is mainly explained by two reasons, both linked to the rise of the Chinese economy. The first reason has to do with the increase in global demand for goods such as copper, driven by the productive dynamics of the Asian giant. The second determinant is related to the decreases in the relative prices of manufactured goods compared to commodities, also partly a consequence of the emergence of this country as a competitor in several major industrial sectors.

Regardless of whether the Prebisch-Singer thesis applies today, there is another worrying element in the commodity market. Commodities exhibit high levels of price volatility. Even researchers such as Sengupta, Narayan and Goswami (2017) show how the dispersion in commodity price indices shows a noticeable increase during the five decades leading up to 2010. This dynamic of wide variability in the prices of goods generates uncertainty in the markets, which is not convenient for the regularity of a nation's fiscal income, the stability of its trade balance, and, in general, for the country's productive structure.

Over time, there has been an increase in volatility and dispersion in the behavior of commodity price indices, with this phenomenon being more pronounced in metals. Consequently, predicting these indices has become a recurrent objective in the literature. As highlighted by Kahraman and Akay (2023), metals play an indispensable role as raw materials in the industry and hold strategic importance in economic development. The prediction of metal prices is crucial for the productive sector and the formulation of effective production policies by countries. The authors conducted an analysis of the prices of aluminum, copper, lead, iron, nickel, tin, and zinc, utilizing annual data spanning from 1990 to 2021. Their findings suggest that exponential smoothing methods prove to be suitable for subsequent forecasting.

Despite the above, it has not been easy for Latin American economies to shift from the production of commodities to the supply of manufactured goods or high-value-added services. Countries such as Colombia, Venezuela, and Peru

exemplify this particular situation, and their dependence on the behavior of exported rents of these goods, which in turn are affected by the variability of international commodity prices, is exposed. Renner and Wellmer (2020) demonstrate how mineral price volatilities, primarily explained by demand conditions, can expose the economies of developing countries that heavily rely on mineral production to significant economic risks. This highlights the need for predictability of such volatility and the behavior of interest indices. Similarly, noteworthy are some works that make interesting efforts to propose empirical frontier applications for predicting metal price indices. For instance, Sohrabi, Jodeiri-Shokri, and Dehghani (2023) forecast coal prices using radial basis function (RBF) and neural networks.

Impact of commodity price behavior on the GDP gap

In this sense, it is reasonable to study the impact of commodity price variation on the output gap and its effect on the unemployment level of the Inca nation in the short term. Alberola et al. (2017) explain that variations in commodity prices can explain the output gap not only from the perspective of their impact on observed GDP, but also in the sense that sustained periods of growth in commodity prices can also affect the trajectory of potential GDP. Using SVAR-type models, Fernández, Schmitt-Grohé, and Uribe (2017) find that commodity price volatility has a significant impact on the economic cycle of countries. The authors consolidate a panel of 138 countries and evaluate the effect of variation in commodity prices (agricultural sector, fuels, and metals) on the output gap, consumption, investment, and international trade for the period 1960–2015. The main finding is that the prices of these goods determine, on average, more than one-third of the fluctuations in the macroeconomic variables referred to. Another relevant conclusion reached by the researchers is that individual commodity prices have greater explanatory power than composite indices in terms of explaining variations in the economic activity of countries. While composite indices explain less than 10% of the economic cycle, individual commodity prices reach figures higher than 30%, as mentioned in the previous paragraph.

Impact of commodity price behavior on unemployment

International commodity prices can affect the unemployment rate of an economy in two ways: 1) indirectly through their impact on the output gap and the subsequent effect of the business cycle on unemployment. 2) directly through the higher income received by export sectors.

Using the Engle-Granger methodology, Eita and Ashipala (2010) find that the output gap is one of the determinants of variations in unemployment levels in

Namibia. Horvath and Zhong et al. (2019) study unemployment dynamics in emerging economies and conclude that increases in external demand can reduce the unemployment rate. The authors represent changes in external demand through the growth rate of U.S. industrial production.

Bayramoğlu (2014) finds a positive and statistically significant impact of increases in agricultural commodity prices on the agricultural employment rate in Turkey. Gaston and Rajaguru (2013) focus on studying the decrease in the equilibrium unemployment rate of the Australian economy over the past decades. The researchers conclude that a significant percentage of this reduction in unemployment is explained by the high values in Australia's terms of trade observed during the period 2000–2018. The authors explain that this improvement in the country's terms of trade can be attributed to the strengthening of commodity exports to China.

Dietsche (2020) argues that the relevant literature on employment and skill development in extractive industries necessitates a comprehensive approach addressing both the supply and demand sides of the labor market, particularly in the context of developing countries where the demand for skills is diffuse and uncertain. Indeed, expectations driven by the global demand for minerals and energy suggest that the extractive sector will provide jobs and skill development opportunities if leveraged through projects funded by companies that drive innovation. Collaboration between governments, businesses, and international partners is crucial to enhance the capacities of citizens in resource-rich countries, especially those residing in proximity to extractive projects. Consequently, the potential outcome could be an improvement in people's living standards, translating into economic growth and development.

Finally, from the literature review, it was found as an interesting finding that the Okun's Law can vary significantly within the same country and between countries, which is confirmed by Ball, Leigh, and Loungani (2017), and adding that this relationship is robust in most advanced countries. However, the effect of a change in production on the unemployment rate varies substantially, so it is key to analyze the particular characteristics of national labor markets, mainly due to the composition of their industries. These results can also be framed in the findings of Briceño, Dávila, and Rojas (2017), who discover that in the case of Ecuador, the results do not fit the Okun model (1962), but they do comply with the data from the world and Latin America. In line with this, Sandrine and Fares (2012) discuss the findings obtained in the estimates of the Okun coefficients for several developed countries and show that there is considerable variation between countries, reflecting the heterogeneity in the response capacity of unemployment to changes in output.

Methodology

Variables, information sources and data

This section provides a detailed description of the methodology implemented in this study, starting with the estimation of potential GDP growth rate and then its difference from the actual GDP growth rate, and subsequently specifying a Structural Vector Autoregressive (SVAR) model that allows us to understand the causal relationships from a theoretical foundation and capture the lagged effects of the Okun's theoretical relationship. The time frame between 1991 and 2019 has been considered based on two criteria. The first one relates to the growth and boom of the mining sector in Peru, which reached levels that consolidated the country as one of the largest exporters of precious metals and minerals of interest to industries in Latin America and the world. On the other hand, it was decided to set the upper limit at the year 2019 since considering the years 2020 to 2022 could distance us from our interest in understanding the transitive impacts derived from a shock in the metal price index on the real economy through the output gap and labor market. This is due to the arrival of the COVID-19 crisis in Latin America in March 2020.

The data used in this study were obtained from the CEPAL information sources for the unemployment rate series and from the Penn World Table version 9.1 constructed by Feenstra, Inklaar, and Timmer (2015) for the GDP series at constant prices in 2011 dollars and IMF Primary Commodity Price Index data base for the Precious Metals Price Index, 2016 = 100, which includes price indexes for gold, silver, palladium, and platinum (Table 1).

Estimation of the product and unemployment gaps

Following Miller's approach (2003), the calculation of the output gap has been considered based on the estimation of the smoothed trend behavior series using the filter developed by Hodrick and Prescott (1997), hereafter Hodrick-Prescott (HP). This decomposition technique is currently the most commonly used to estimate Okun's Law, with the HP filter applied to the observed production variable or real cyclical GDP, to find its trend component and theoretically determine its optimal levels (Rodríguez and Peredo-Rodríguez, 2007, p.63). Therefore, the time series to be decomposed has an implied trend component denoted by τ_t and a cyclical component c_t ; $\forall y_t = \tau_t + c_t$, : thus:

$$\min \sum_{t=1}^T (y_t - \tau_t)^2 + \lambda \sum_{t=1}^{T-1} [(\tau_{t+1} - \tau_t) - (\tau_t - \tau_{t-1})]^2 \quad (3)$$

The trend component of a series is the one that minimizes Eq. (1). The criterion for determining the optimal value of λ is generally arbitrary; however, if the series

Table 1 Descriptive statistics 1991–2019

	GDP Gap	Unemployment gap	Metals price index
Mean	-60.11046	0.006670	63.37944
Median	-161.8355	-0.015915	45.10854
Maximum	10873.71	0.553298	136.6160
Minimum	-9153.924	-0.360930	24.33217
Std. Dev	5942.600	0.220638	38.86618
Skewness	0.142788	0.501265	0.463735
Kurtosis	1.869239	2.769610	1.638451
Jarque–Bera	1.586870	1.234502	3.166355
Probability	0.452289	0.539425	0.205322
Sum	-1683.093	0.186754	1774.624
Sum Sq. Dev	9.53E + 08	1.314392	40785.66
Observations	28	28	28

Source: own elaboration based on data from CEPAL and Penn World Table version 9.1 and price index retrieved from IMF Primary Commodity Price Index data base for the Precious Metals Price Index, 2016 = 100, which includes price indexes for gold, silver, palladium, and platinum

decomposition process is carried out based on the specific method of estimating potential product growth using the filter, Tvrdon (2012) recommends using an optimal value between $\lambda = 1600$ and $\lambda = 100$ for quarterly data over a year. Specifically, in the present work, the decomposition process is carried out using a value of $\lambda = 1600$.

According to the National Institute of Statistics and Informatics (INEI) of Peru and the Central Reserve Bank of Peru, in 2014 the lowest unemployment rate of the last two decades was evidenced with 5.4%. Since 2016, the fragility of the Peruvian economy has been hit by the drop in international prices of primary goods (including minerals), which fully affects the loss of formal jobs and the growing informality (see Figs. 1 and 2). However, a sustained growth of capital stock is observed for the analysis period, which translates into an increase in the growth of the Peruvian economy for the period prior to the commodity price crisis (see Fig. 1).

It is believed that the current period of economic expansion in the country has been driven by domestic demand since the turn of the century. This has been sustained by an increase in private final consumption, government consumption, and recovery of investment in the component of real fixed capital stock, both public and private. As a result, labor-intensive manufacturing, agricultural, and mining activities have been promoted in the last decade, according to the National Institute of Statistics and Informatics (INEI, 2019). This growth and increase in productivity have led to a significant decrease in the unemployment rate. Here's a possible translation:

The objective is to capture the expansive or contractive dynamics of the business cycle, as suggested by Grant

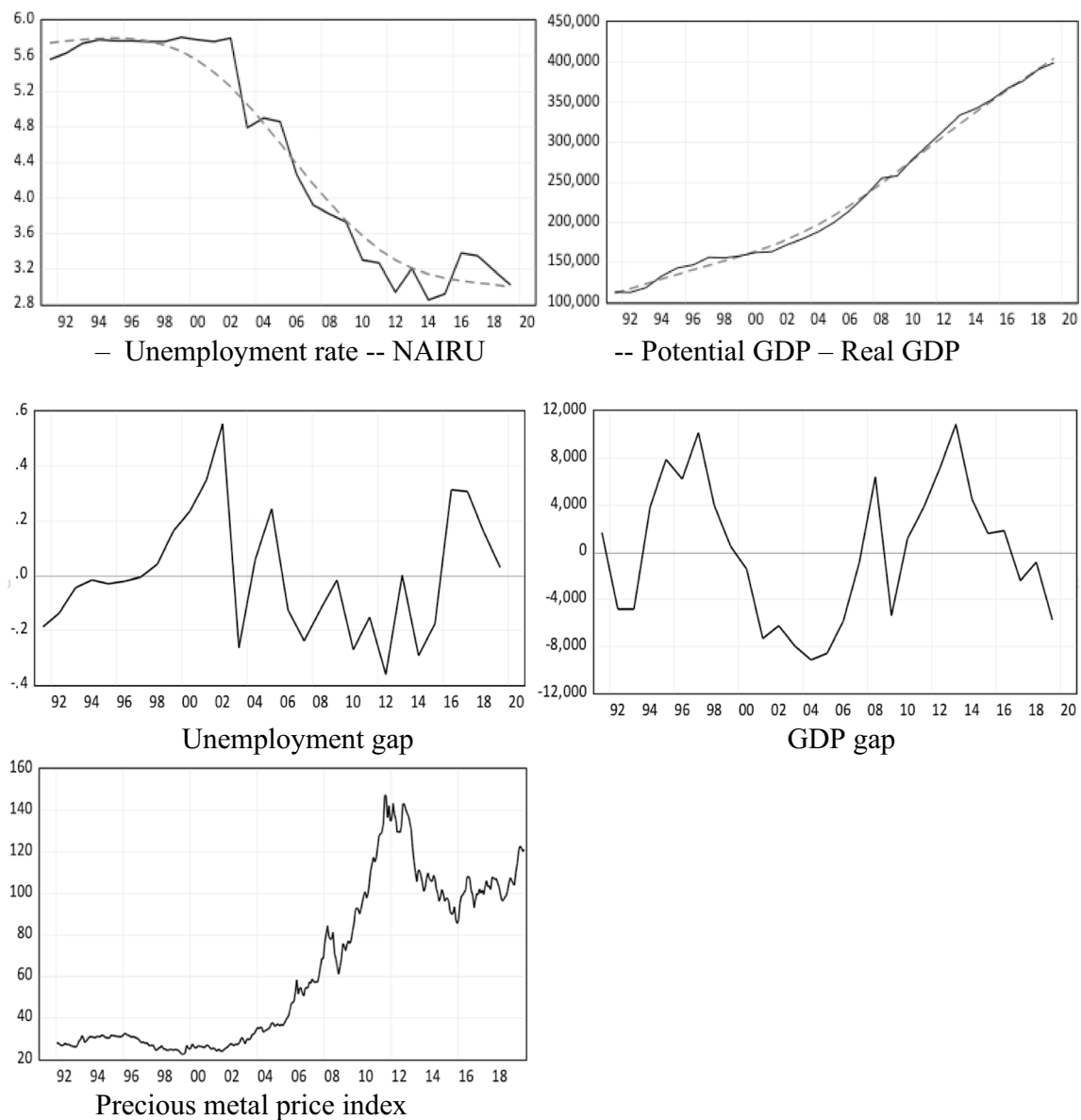


Fig. 1 Potential GDP, real GDP, Output Gap, Unemployment Gap and metal price index 1991–2019.

Source: own elaboration based on data from CEPAL and Penn World Table version 9.1 and price index retrieved from IMF Primary Com-

modity Price Index data base for the Precious Metals Price Index, 2016=100, which includes price indexes for gold, silver, palladium, and platinum

(2002). We believe that these dynamics are transitively related to the behavior of international prices of precious metals, which is a sector of significant importance for the Peruvian economy. This approach is fundamental since it allows for the analysis of cyclical movements over time, which implies a greater or lesser use of productive resources, leading to the possibility of proposing a series of potential output figures that are consistent with an approximate level of optimal use of productive resources.

Based on the above, this study also includes the variation of the unemployment gap as the difference between the

unemployment rate in periods t and $t-1$, all within a structural vector autoregressive system, whose identification and specification are described below in subsection 3.2. The variables analyzed in the present study are presented below.

Identification of a structural VAR model with a system of $I(1)$ stationary variables

Regarding the proposed objective, Cardona-Arenas and Osorio (2016), based on Bande and Karanassou (2013), suggest that the development of an interactive system of labor

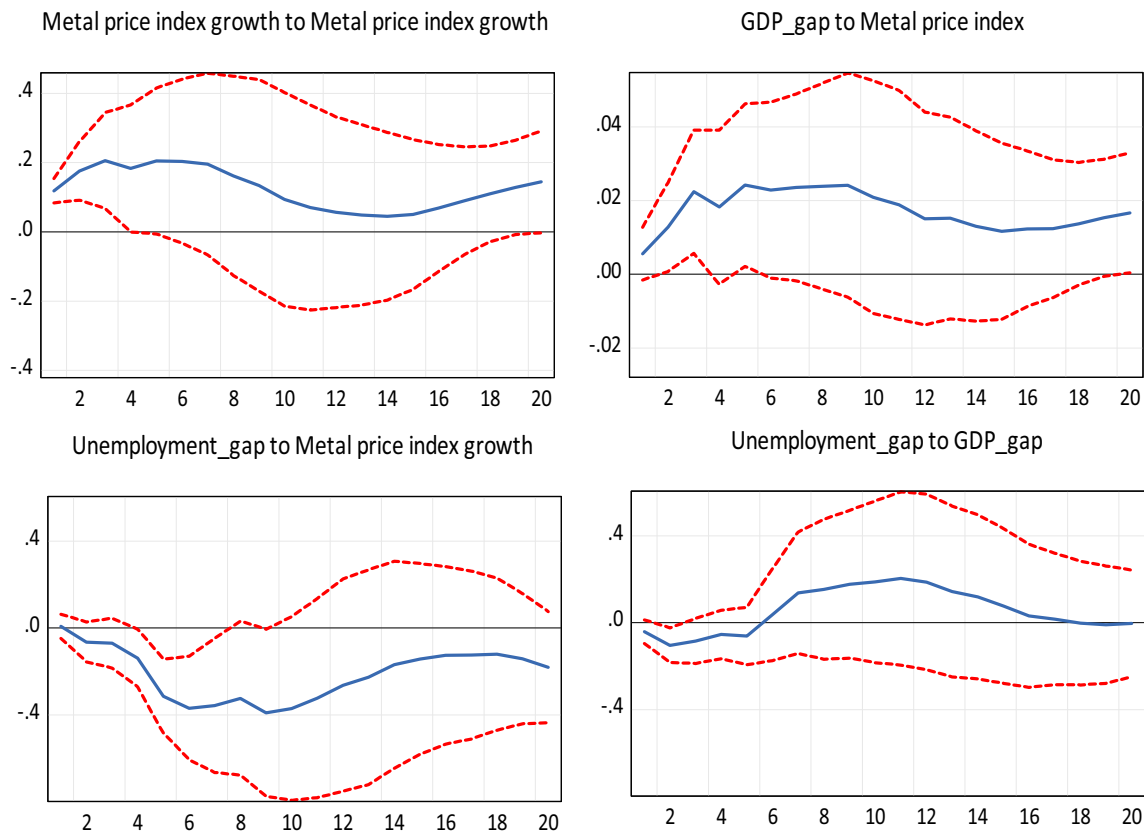


Fig. 2 Generalized Accumulative response to structural SVAR one S.D innovations.

Source: Own elaboration based on SVAR estimation and data from CEPAL and Penn World Table version 9.1 and price index retrieved

market equations is preferable to interpret unemployment dynamics. Additionally, Cardona-Arenas and Sierra-López (2020) and Cardona-Arenas et al. (2023) propose a dynamic multi-equational system highlighting the usefulness of vector autoregressive analysis, in the former case to understand the dynamic and lagged relationships in highly volatile variables, and in the latter case to impose a transitive analysis to analyze spillover effects channels among the highly volatile variables system. Therefore, the present work is based on these methodological principles but also proposes a specific order in which effects are transmitted between variables. Thus, a dynamic multi-equational system is specified based on the S-VAR methodology proposed for a k -dimensional vector of $n \times 1$ endogenous variables denoted by X_t , which are non-cointegrated of order I (1) and whose behavior can be represented through the following structural form:

$$A_0 X_t = B(L)X_{t-1} + \varepsilon_t \quad (4)$$

$$X_t = A_0^{-1} B(L)X_{t-1} + A_0^{-1} \varepsilon_t \quad (5)$$

from IMF Primary Commodity Price Index data base for the Precious Metals Price Index, 2016=100, which includes price indexes for gold, silver, palladium, and platinum. Note: Cholesky ordering Metal price index growth- GDP gap – Unemployment gap

$$X_t = \tilde{B}(L)X_{t-1} + \tilde{A}\varepsilon_t \quad (6)$$

The vector of variables X_t constitutes a (3×1) matrix of stationary deseasonalized variables, $X_t = [y_t, im_t, u_t]$. $B(L)$ is an $n \times n$ polynomial matrix in the lag operator (L), of the form $B(L) = \sum_{i=1}^k BL^i = B_1L + B_2L^2 + \dots + B_kL^k$, and ε_t is an $n \times 1$ vector of structural residuals that is characterized as a white noise vector, with zero mean, no serial correlation, and a covariance matrix that is constant over time. A_0 is a triangular $n \times n$ matrix that describes the contemporaneous relationships between the endogenous variables and allows for the imposition of restriction conditions in the model to control for contemporaneous effects. Therefore, to impose restrictions, we follow the approach of estimating a recursive system for identifying the system of equations, assuming that the structural shocks must be orthogonal to each other. To simplify the exposition, we assume the following recursive S-VAR form:

$$Metal_index_t = \dots + \tilde{a}_{11}\varepsilon_{mt} \quad (7)$$

$$GDP_gap_t = \dots + \tilde{a}_{21}\varepsilon_{mt} + \tilde{a}_{22}\varepsilon_{yt} \quad (8)$$

$$Unemployment_gap_t = \dots + \tilde{a}_{31}\varepsilon_{mt} + \tilde{a}_{32}\varepsilon_{yt} + \tilde{a}_{33}\varepsilon_{ut} \quad (9)$$

In this recursive specification, it is assumed that $Metal_index_t$ is the most exogenous variable within the system of variables, since to describe the relationship described by Okun, we must assume that the dynamics of the economic cycle influence this variable and therefore are only affected by a shock in itself (which can be of an exogenous nature). On the other hand, the most endogenous variable in the system will be the unemployment gap, to be consistent with the relationship described by Okun, and the GDP gap is the variable that allows the shock to be transmitted from the metal index to unemployment gap, as shown below:

- ε_{mt} affects contemporaneously all variables
- ε_t affects contemporaneously to GDP_gap_t and $Unemployment_gap_t$
- ε_{ut} affects contemporaneously to u_t but not to y_t e Im_t

It is important to note that some studies have identified commodity prices as a more exogenous variable. This is evident in the work of Zhong et al. (2019), who explore the relationship between international prices of non-ferrous metals and industrial economic variables in China. The findings suggest that metal prices (considered the most exogenous variable in the system) positively impact employment. Shocks in non-ferrous metal prices emerge as a significant force in economic fluctuations, with their explanatory power intensifying after the global financial crisis.

The effect of the shocks induced in the system of equations variables, according to the imposed restrictions, is measured by means of structural impulse-response functions. For the correct specification of the model, it is crucial to perform the respective unit root tests to check the integration order

of the I (1) series. Therefore, the series under analysis in this research article are stationary in first differences, and compliance with the condition of no serial autocorrelation is ensured. This is done by testing for unit roots using the Dickey-Fuller (1981) and Phillips-Perron (1988) tests (see Table 2), which allows us to conclude that the series of output gap, Metal price index, and unemployment gap are integrated of order (1). Likewise, in appendix (B) of the study, the serial autocorrelation LM-test shows that for the estimated third-order SVAR model, errors are not correlated over time. The model estimation order is proposed based on this criterion and adjusted from the information provided by the Akaike information test, which initially yields an order of four but, in different autocorrelation tests, suggests a preferred order of three lags, see Table 3 in Appendix (A) at the end of the study. Next, the level and first-difference results of the variables are presented.

Results and discussion

Impulse response functions measure the effect of a positive shock in the explanatory variables, given the ordering imposed in the short-term restrictions, the reactions in period $t=1$ after the shock, and up to 20 periods later, demonstrating that the theoretical relationship between unemployment and the output gap is consistent with the theory of Okun's Law since the sign is as expected (see Fig. 2). In detail, we can observe that a positive shock in the metal price index produces a positive and statistically significant effect on the output gap and produces a reduction in the variation of unemployment, a statistically significant effect and different from zero effect up to two periods after the shock. The evidence is solid to conclude that Okun's Law is applicable to the Peruvian economy when a shock in the metal

Table 2 Test unit root test Dickey-Fuller y Phillip-Perron

	Augmented Dickey-Fuller test				Phillip-Perron test statistic				
	ADF in levels		ADF first differences		PP in levels		PP first differences		Integration Order
<i>Trend and Intercept</i>	<i>T-Stat.</i>	<i>Prob.</i>	<i>T-Stat.</i>	<i>Prob.</i>	<i>T-Stat.</i>	<i>Prob.</i>	<i>T-Stat.</i>	<i>Prob.</i>	
Metal price index	-2.75	0.2241	-2.54	0.015	-0.626	0.848	-2.67	0.009	I(1)***
GDP gap	-2.13	0.5027	-5.85	0.000	-2.24	0.447	-5.59	0.000	I(1)***
Unemployment gap	-3.50	0.0587	-5.75	0.004	-3.45	0.064	-14.72	0.000	I(1)***

Source: own elaboration based on data from CEPAL and Penn World Table version 9.1 and price index retrieved from IMF Primary Commodity Price Index data base for the Precious Metals Price Index, 2016=100, which includes price indexes for gold, silver, palladium, and platinum. Note: The symbols *, **, and *** correspond to the level of statistical significance: 10%, 5%, and 1%, respectively. According to the results, the variables are integrated in order 1 for a significance level of 5% in the ADF-Augmented Dickey-Fuller test, variables adjusted by seasonal and trend decomposition using Loess. STL stands for Seasonal-Trend Decomposition

price index occurs in the short term. This study is possibly the first work in literature that examines this relationship in the country using the proposed methodology. Additionally, as shown in Fig. 2, a positive shock in investment results in a statistically significant increase in the output gap for the first three periods following the shock.

According to statistical evidence, it is possible to highlight the finding consistent with Okun's Law, in which shock in some variable that affects the output gap reduces the unemployment gap. Therefore, the unemployment rate reacts downward as proposed by Okun's Law, fulfilling the assumption. From the above, Table 5 in Appendix (D), it is possible to observe the statistically significant and different from zero effects that an innovation in the output gap exerts on unemployment variation during the first two periods after the shock. Additionally, a positive shock in the metal price index results in a significant response in the output gap, highlighting the strong connection between Peru's economic growth and the mining exports sector.

As evidenced in Fig. 3, a variance decomposition analysis is performed, see Table 4 in Appendix (C), which reflects the proportion of movements in price metal index, the output gap and unemployment gap that are due to shocks to itself and to shocks derived from other variables in the model. From the percentage of variation in the output gap, it can be said that it does not behave as a completely exogenous variable to the other variables incorporated in the model, since although in period 1 it is explained 100% and stabilizes from there until period 16, with a percentage of 60.96%, which in practice is consistent with the endogeneity characteristic of the business cycle.

The percentage of GDP gap variance explained by the metal price index is 10.06% in the first period after the shock, increasing to 35.60% by the 16th period after the shock. On the other hand, in period 1 after the shock, the variation in the unemployment gap is explained 0.0% by itself, but by period 16 when the shock effect stabilizes, it is explained by 51.20% by the growth of the price metal index. Furthermore, the percentage of variation in the output gap explained by the GDP gap is 9.08% in the first period after the shock and 29.08% in the period 16 after the shock induced.

Finally, it is noteworthy that both the output gap and the unemployment gap are significantly influenced by a shock to the metal price index, with a considerable amount of variance explained. This shock has significant effects throughout the real economy and labor market, aspects that can easily be overlooked and highlight the high dependence of the Peruvian economy on the performance of the mining sector, which in turn is affected by international prices of precious metals and other primary mining goods.

The generalized Cholesky factor methodology for structural shocks, applied to the output gap in Peru,

enables the historical decomposition process of variance (HDV). This process allows for calculating the contribution of different shocks to the accumulated output gap, by orthogonalizing the shocks through the S-VAR specification. It is worth noting that, unlike the method suggested by Diebold and Yilmaz (2012), structural estimation does not require shock orthogonalization. HDV can be interpreted as a tool for assessing the relative importance of different shocks in explaining the variability of the variables in the model. Furthermore, HDV can also help identify how different shocks are correlated with each other, which can be useful in better understanding the interaction between different variables in the model. In this exercise, we consider only three types of shocks: growth of metal price index, output gap variations in unemployment gap.

The Fig. 5 in Appendix I of this document presents the graph corresponding to the historical variance decomposition. In general, it can be observed that the shock in the metal price index explains most of the variance in the GDP gap and Unemployment gap variables, highlighting a significant dependency of aggregate economic activity on the external behavior of the precious metals market and their prices. As a result, there is an interest on the part of central planners to improve their terms of trade (relative price of exports to imports) to ensure that the positive effects resulting from, for instance, increases in international prices of precious metals counterbalance potential currency depreciations (if import demand is inelastic). Currency depreciation could lead to a significant deterioration in Peru's trade balance. Sobrino (2011) points out, the continuous improvement of the terms of trade has a direct implication of a permanent increase in the standard of living, stemming from heightened productivity due to a subsequent more efficient reallocation of resources. The author emphasizes that empirical evidence demonstrates that an improvement in the terms of trade translates into an increase in per capita GDP. In a broader context of countries, the outcomes for each nation will essentially depend on the economic export sector to which they are most reliant. Generally, these results may be applicable to economies primarily exporting precious metals. Nevertheless, we aim to expand this research to establish a panel approach that incorporates more primary-exporting countries, particularly those highly dependent on foreign exchange earnings from the export of precious metals.

The results of this research provide evidence of a close and inverse relationship between the unemployment gap and the output gap. This relationship is significant when an exogenous shock in the growth variable of the metal price index occurs, demonstrating that there is external dependence of economic performance on changes in the international market prices of commodity metals. As stated by Renner and Wellmer (2020), the author argues that the global metal

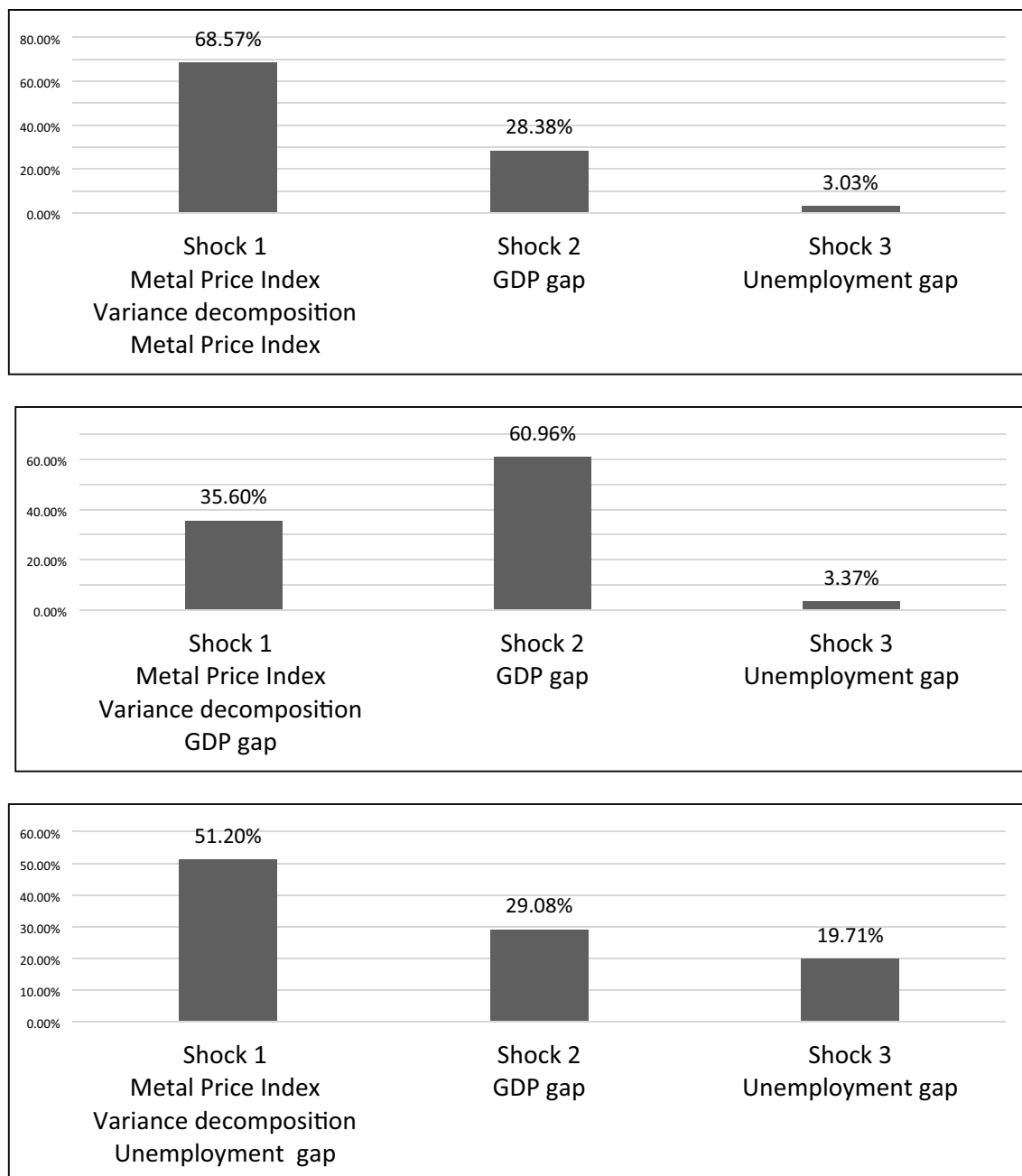


Fig. 3 Variance decomposition analysis using structural SVAR factors. Source: Own elaboration based on SVAR estimation and data from CEPAL and Penn World Table version 9.1 and price index retrieved from IMF Primary Commodity Price Index data base for the Precious

Metals Price Index, 2016=100, which includes price indexes for gold, silver, palladium, and platinum. Note: Cholesky ordering Metal price index growth- GDP gap – Unemployment gap

price market predominantly operates as a buyers' market. In this context, prices and their volatility are primarily driven by changes in demand patterns rather than being constrained by supply limitations. Our research has also considered the impact of parameters on metal prices in various countries. According to Raddatz (2008), fluctuations in economic growth in Latin American countries are strongly influenced

by external shocks, particularly those related to terms of trade. In such cases, erratic fiscal or monetary policies may exacerbate these impacts Loayza and Raddatz (2006). This implies that, when considering the influence of parameters on metal prices, it is essential to incorporate variables that capture the dynamics of global demand, such as overall global economic growth or market terms of exchange,

instead of solely focusing, for example, on the growth rates of specific producing countries as determinants of precious metal prices. The aforementioned validates the methodological proposal of this study and its results.

From this analysis of the macroeconomic determinants of variation in the unemployment and output gaps, it can be inferred that the process of adjusting the output gap has an impact on reducing the variation of the unemployment gap when increases in international prices of precious metals take place. Based on the results, it can be inferred that approximately 61% of the variation in the production gap is explained by its history, which shows signs of inertial behavior explained by high memory in the series of this variable. This finding is consistent with Silvapulle, Moosa, and Silvapulle (2004), who support the proposition that the production-unemployment relationship in gaps represented by Okun's law is asymmetric and not necessarily constant over time. Therefore, the results of this study are of great interest since they demonstrate that short-term effects on the production gap from changes in the precious metals price index are related to the labor market, specifically with the unemployment gap, in which case a positive product gap would promote a reduction in the unemployment gap, leading the Peruvian economy to potential levels of unemployment and production.

However, it is known that the cyclical variation of unemployment is quantitatively different from job creation, depending on both the periods considered and the level of industrial diversification of the economy. Therefore, it is expected that the transitive relationship identified in this study should be further analyzed in consistency with the theoretical explanation of the asymmetry in the Okun coefficient. It is important to mention that autoregressive vector models contribute valuable insights to the analysis of the Okun's Law, allowing us to consider the lagged effects accumulated in the output gap calculated from the estimation of potential GDP and the unemployment gap variation. This vectorial analysis allows to track the transmission of asymmetric shocks over time simulated after the induced shock. Additionally, the analysis of impulse response functions allows to isolate the effect of a specific shock on the output gap and observe its effect on the unemployment dynamics, as evidenced in Melguizo's (2017) work, who estimates a panel VAR model to analyze Okun's Law in the Spanish provinces and finds evidence of the relationship described by Okun (1962).

Specifically, the impulse response functions obtained from the proposed multi-equational system in this study indicate that the theoretical relationship between unemployment and the output gap is consistent with Okun's Law for Peru. Thus, the hypothesis of an inverse relationship between unemployment variation and the output gap in the short term is verified. Interestingly, the shock introduced

into the system of equations is applied to the variable of growth in the index of precious metal prices. There would be no guarantee that this shock would have an impact on the variables of the product gap and the unemployment gap if there were little or no export dependence on precious metals traded in international markets. This study demonstrates that the Peruvian real economy continues to be strongly linked to the income generated by mining exports.

The increased reliance on economic performance in the mining sector undoubtedly results in a negative externality, elevating the susceptibility to exchange rate volatility in response to significant fluctuations in the international prices of the exportable commodity. This observation aligns with the findings of Cardona et al. (2020) in the context of Colombian energy commodities. The authors specify that the Colombian economy's dependence on oil exports implies that the price of oil influences the behavior of the nominal exchange rate in Colombia. However, they note that the nominal exchange rate does not influence the international price of oil, indicating a unidirectional causality. Consequently, it is inferred that the dynamics of the exchange rate are responsive to external factors, such as changes in international prices of precious metals, as exemplified in the case of Peru.

Furthermore, it is important to mention that the results obtained in this study are consistent with the findings in more developed economies, where other studies conclude that this law cannot be confirmed mainly in developing countries or those with little industrial diversification and due to the complexity of the structural changes presented in their labor markets. Therefore, for future extensions of this research, it is proposed to continue with the approach of various works that propose cycle and trend decomposition analysis as a means to understand the compliance of Okun's Law in countries with strong economic expansion processes such as exhibited by Peru during much of the analysis period. Among these works, Gordon (1984), Freeman (2001), Holmes and Silverstone (2006) can be mentioned.

The results suggest that the Okun curve (1962) lacks strong support in the data and is rather a particularity of some developed countries. As a result, this study opens up a debate on whether compliance with Okun's Law for developing countries should be empirically analyzed, considering sectors that promote foreign currency inflows, job creation, or capital-intensive sectors related to national production chains. In this sense, this study presents an analysis of an origin variable and its transitive effects on the performance of the Peruvian economy, examined from the theoretical perspective of Okun's Law. The aim is not to validate or refute the theory, but rather to provide context. We believe that this objective has been achieved with this study and it contributes to the analysis of determinants of economic performance in emerging or developing economies.

Conclusions

Okun's Law aims to establish the relationship between potential output, cyclical growth of output, and the variation in unemployment rate with respect to its trend level. Okun stressed the importance of measuring this gap to understand the effect of production level changes on unemployment reduction and vice versa.

This study examines the sensitivity of the Peruvian labor market to changes in the production gap generated by shocks in international primary goods prices, particularly precious metals, using the SVAR estimation results and other facts. While most existing literature rejects the presence of Okun's Law for Latin American countries, this study finds evidence of its existence for Peru during the period 1992–2019. This study is innovative in identifying realistic transitive effect channels with the conditions of Latin American economies. The proposed model aims to overcome previous limitations by exploring a more complete and realistic transitive approach. Based on the identification of macroeconomic fundamentals, it can be inferred that the adjustment process

of the variation in the unemployment and production gaps occurs through adjustments in the international markets of precious metals, reflecting Peru's economic performance regarding the production and unemployment gaps.

The study suggests that the Peruvian economy has increased its capital-labor ratio in the mining sector in a slow but steady structural change process, and therefore, the unemployment gap responds decisively to significant changes in economic growth in Peru. This study demonstrates that the Peruvian real economy continues to be strongly linked to the income generated by mining exports.

The need to analyze other leading indicators, such as total factor productivity, the cost, and marginal return of capital, is highlighted for future research. This study opens the discussion window regarding the impact of the structural change process on compliance with Okun's Law, an aspect that will be evaluated in future research and other Latin American countries to corroborate this hypothesis.

Appendix A

Table 3 Lag inclusion test

Lag	LogL	LR	FPE	AIC	SC	HQ
0	74.33759	NA	3.06e-07	-6.485235	-6.336457*	-6.450188
1	83.98392	15.78491	2.92e-07	-6.543993	-5.948879	-6.403802
2	91.06526	9.656373	3.68e-07	-6.369569	-5.328120	-6.124235
3	97.13638	6.623040	5.56e-07	-6.103308	-4.615522	-5.752830
4	122.8556	21.04300*	1.67e-07*	-7.623237*	-5.689117	-7.167617*
5	130.6435	4.247932	3.54e-07	-7.513044	-5.132588	-6.952280

* indicates lag order selected by the criterion

LR: sequential modified LR test statistic (each test at 5% level)

FPE: Final prediction error

AIC: Akaike information criterion

HQ: Hannan-Quinn information criterion

Source: Own elaboration based on data from CEPAL and Penn World Table version 9.1 and price index retrieved from IMF Primary Commodity Price Index data base for the Precious Metals Price Index, 2016=100, which includes price indexes for gold, silver, palladium, and platinum

Appendix B

Structural VAR Estimates

Sample (adjusted): 1997 2019

Included observations: 23 after adjustments

Estimation method: Maximum likelihood via Newton–Raphson (analytic derivatives)

Convergence achieved after 14 iterations

Structural VAR is just-identified

Model: $e = \Phi u$ where $E[uu'] = I$

F =

C(1)	0	0		
C(2)	C(4)	0		
C(3)	C(5)	C(6)		
	Coefficient	Std. Error	z-Statistic	Prob
C(1)	0.14848	0.02189	6.78232	0.000
C(2)	0.01065	0.00345	3.08244	0.002
C(3)	-0.21066	0.03837	-5.48953	0.000
C(4)	0.01476	0.00217	6.78232	0.000
C(5)	-0.07639	0.01952	-3.91361	0.000
C(6)	0.07645	0.01127	6.78232	0.000
Log likelihood	92.94310			
Estimated S matrix:				
0.08885	0.07269	-0.02966		
-0.00618	0.01644	-0.00051		
0.06036	-0.01321	0.11768		
Estimated F matrix:				
0.14848	0.00000	0.00000		
0.01065	0.01476	0.00000		
-0.21066	-0.07639	0.07645		

Source: Own elaboration based on data from CEPAL and Penn World Table version 9.1 and price index retrieved from IMF Primary Commodity Price Index data base for the Precious Metals Price Index, 2016 = 100, which includes price indexes for gold, silver, palladium, and platinum

Appendix C

Table 4 Serial autocorrelation test of VAR residuals, LM-Test

Null hypothesis (H0): There is no serial autocorrelation in lag (n)						
Lag(n)	LRE* stat	df	Prob	Rao F-stat	df	Prob
1	7.403636	9	0.5952	0.816931	(9, 12.3)	0.6118
2	9.572204	9	0.3862	1.138069	(9, 12.3)	0.4064
3	15.15086	9	0.0869	2.198057	(9, 12.3)	0.0998
4	13.37918	9	0.1462	1.820144	(9, 12.3)	0.1627
5	7.642290	9	0.5706	0.850156	(9, 12.3)	0.5878

Source: Own elaboration based on data from CEPAL and Penn World Table version 9.1 and price index retrieved from IMF Primary Commodity Price Index data base for the Precious Metals Price Index, 2016 = 100, which includes price indexes for gold, silver, palladium, and platinum

Appendix D

Table 5 Variance decomposition

	Metal Price Index Shock 1	GDP gap Shock 2	Unemployment gap Shock 3
Metal Price Index			
t=1	100	0	0
t=2	93.81	4.19	1.99
t=15	68.27	28.67	3.04
t=16	68.57	28.38	3.03
GDP gap			
t=1	10.06	89.93	0.00
t=2	22.36	75.55	2.08
t=15	35.66	60.96	3.37
T=16	35.60	60.96	3.37
Unemployment gap			
t=1	0.33	9.80	89.86
t=2	17.54	18.63	63.81
t=15	51.05	29.17	19.76
t=16	51.20	29.08	19.71

Source: Own elaboration based on SVAR estimation

Appendix E

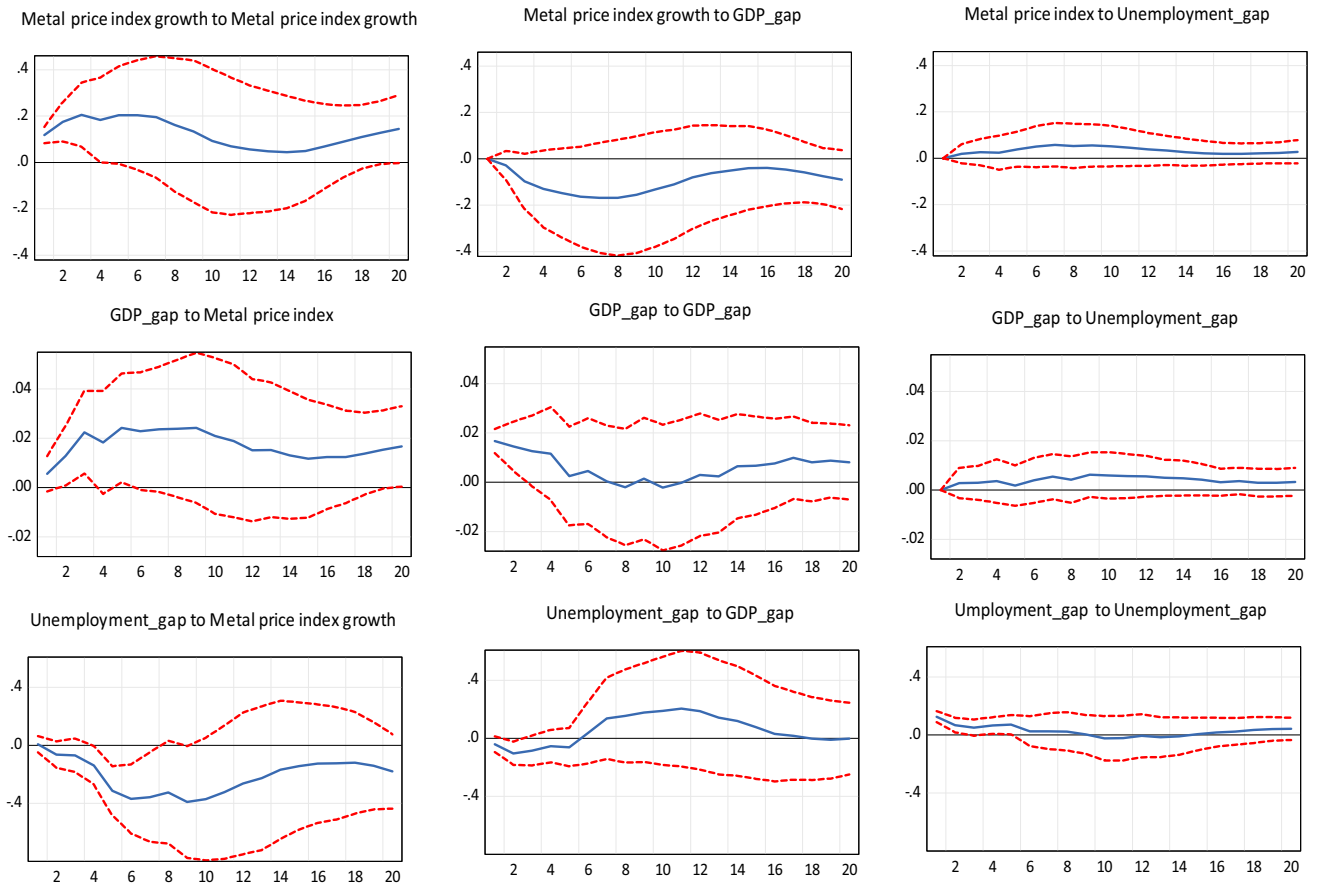


Fig. 4 Cumulative FIR to one S.D structural innovations. Source: Own elaboration based on SVAR estimation – Cholesky ordering Metal price index growth- GDP gap – Unemployment gap

Appendix G

Table 6 Test de Causalidad de Granger

Null Hypothesis:	Obs	F-Statistic	Prob
Unemployment gap does not Granger Cause GDP gap	27	2.50405	0.1048
GDP gap does not Granger Cause Unemployment gap		0.63066	0.5416
Metal price index does not Granger Cause GDP gap	26	0.67175	0.5215
GDP gap does not Granger Cause Metal price index		2.03723	0.1554
Meta price index does not Granger Cause Unemployment gap	26	1.00936	0.3815
Unemployment gap does not Granger Cause Metal price index		0.12172	0.886

Source: Own elaboration based on SVAR estimation. Note: Test for 2 Lag included

Appendix H

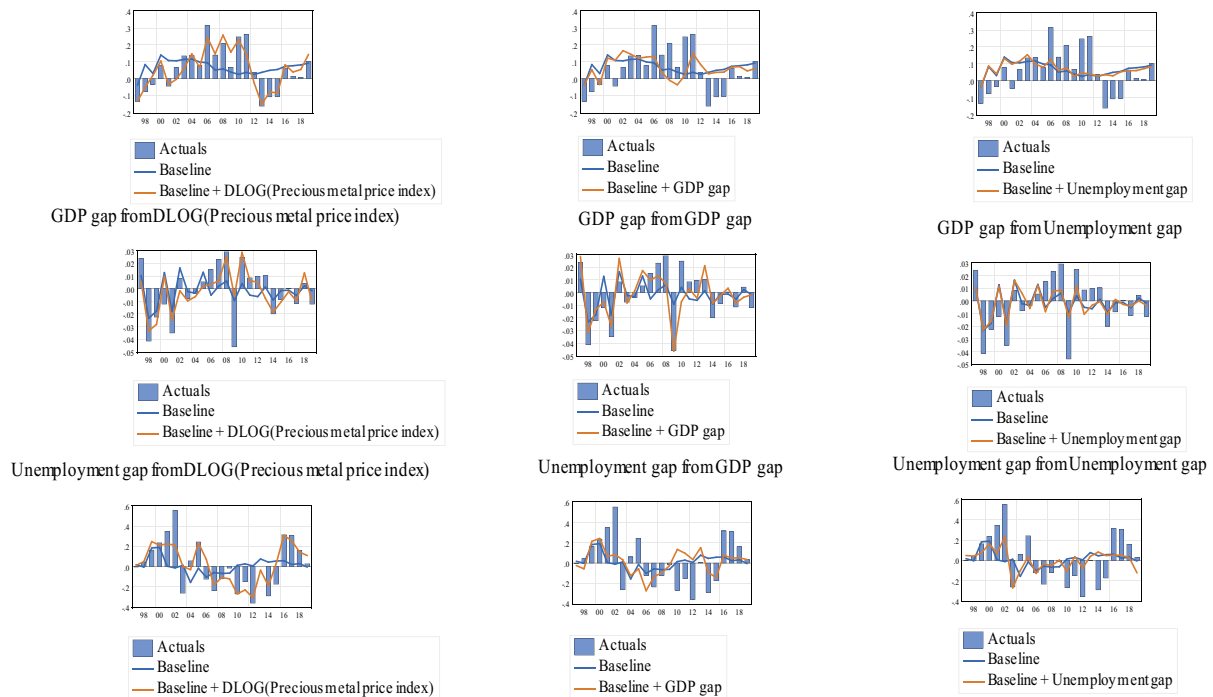
Table 7 Cointegration test

Sample (adjusted): 1991—2019				
Included observations: 26 after adjustments				
Trend assumption: Linear deterministic trend				
Series: BRECHA_PRODUCTO BRECHAU PRECIOUS METALS PRICE_INDEX				
Lags interval (in first differences): 1 to 1				
Unrestricted Cointegration Rank Test (Trace)				
Hypothesized		Trace	0.05	
No. of CE(s)	Eigenvalue	Statistic	Critical Value	Prob.**
None	0.475665	24.02015	29.79707	0.1996
At most 1	0.239038	7.233928	15.49471	0.5506
At most 2	0.005043	0.131445	3.841466	0.7169
Trace test indicates no cointegration at the 0.05 level				
* denotes rejection of the hypothesis at the 0.05 level				
**MacKinnon-Haug-Michelis (1999) p-values				
Unrestricted Cointegration Rank Test (Maximum Eigenvalue)				
Hypothesized		Max-Eigen	0.05	
No. of CE(s)	Eigenvalue	Statistic	Critical Value	Prob.**
None	0.475665	16.78623	21.13162	0.1823
At most 1	0.239038	7.102483	14.26460	0.4771
At most 2	0.005043	0.131445	3.841466	0.7169
Max-eigenvalue test indicates no cointegration at the 0.05 level				
* denotes rejection of the hypothesis at the 0.05 level				
**MacKinnon-Haug-Michelis (1999) p-values				
Unrestricted Cointegrating Coefficients (normalized by b'S11*b=I):				
BRECHA_PRODUCTO	BRECHAU	PRECIOUS_METALS_PRICE_INDEX		
-0.000131	-6.774820	-0.008425		
0.000186	-1.238493	-0.003889		
-7.53E-05	-0.003443	0.029456		

Source: Own elaboration based on data from CEPAL and Penn World Table version 9.1 and price index retrieved from IMF Primary Commodity Price Index data base for the Precious Metals Price Index, 2016=100, which includes price indexes for gold, silver, palladium, and platinum

Appendix I

DLOG(Precious metal price index) from DLOG(Precious metal price index) DLOG(Precious metal price index) from GDP gap DLOG(Precious metal price index) from Unemployment gap



Historical Decomposition of Variance (HDV). Source: Own elaboration based on SVAR estimation

Authors contribution All authors contributed to the study conception and design.

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Data Availability The data is available upon written request addressed to the authors.

Declarations

Financial interest The author declare they have no financial interests, the authors have no relevant financial or non-financial interests to disclose.

References

- Neffa JC, Albano J, Ghio RL, Pérez PE, Salas J, Toledo F (2007) Teorías económicas sobre el mercado de trabajo II. Neoclásicos y nuevos keynesi
- Apergis N, y Rezitis A (2003) An examination of Okun's law: evidence from regional areas in Greece. *Appl Econ* 35(10):1147–1151. <https://doi.org/10.1080/0003684032000066787>
- Attfield CL, y Silverstone, B. (1998) Okun's law, cointegration and gap variables. *J Macroecon* 20(3):625–637. [https://doi.org/10.1016/S0164-0704\(98\)00076-7](https://doi.org/10.1016/S0164-0704(98)00076-7)
- Ball L, Leigh D, y Loungani, P (2017) Okun's Law: Fit at 50?. *J Money Credit Bank* 49(7):1413–1441. <https://doi.org/10.1111/jmcb.12420>
- Bande R, y Karanassou M (2013) The Natural Rate of Unemployment Hypothesis and the Evolution of Regional Disparities in Spanish Unemployment. *Urban Stud* 50(10):2044–2062. Recuperado de <http://www.jstor.org/stable/26144590>
- Blanchard O, Gali J (2008) Labor markets and monetary policy. Cambridge: Nber working paper series. <https://doi.org/10.1257/mac.2.2.1>
- Boďa M, y Považanová M (2015) Gender Asymmetry in Okun's Law in the Four PIGS Countries. *Procedia Economics and Finance*, 30: 111–123. [https://doi.org/10.1016/s2212-5671\(15\)01260-5](https://doi.org/10.1016/s2212-5671(15)01260-5)
- Briceño MB, Dávila G, y Rojas, M (2017) Estimación de la Ley de Okun: evidencia empírica para Ecuador, América Latina y el Mundo. *ReVista Económica* 1(1):35–45 (Recuperado de <https://revistas.unl.edu.ec/index.php/economica/article/view/209>)
- Cardona Arenas CD, Osorio Barreto D, Toro Díaz J, Teran Velazco CA, Redondo Ramírez MI, Díaz Restrepo CA, ... & Cuellar Marquez N (2021) Financial economics: Dynamic analysis of macro fundamental variables and risk cases, Colombia and Lima (Peru). Retrieved from: <https://repository.unilibre.edu.co/handle/10901/19757>

- Cardona-Arenas CD, y Osorio, D (2016) Aproximación al análisis de disparidades regionales en términos del desempleo: una revisión de literatura. *Revista de Economía y Administración*, 13(1), 131–151. Recuperado de <https://revistas.uao.edu.co/ojs/index.php/REYA/article/view/13>
- Cardona-Arenas CD, Sierra-Suárez LP (2020) Impacto de la política monetaria en el equilibrio del mercado de trabajo: países de la Alianza del Pacífico. *Revista Finanzas y Política Económica* 12(2):491–521
- Cardona-Arenas CD, Gómez-Gómez R, Morales-Zuluaga E (2023) COVID-19 and its short-term informational impact on the stock markets of the Pacific Alliance countries. *SN Busi Econ* 3(5):1–23. <https://doi.org/10.1007/s43546-023-00469-6>
- Central Reserve Bank of Peru (2021) Memory Archive 1991–2001. Lima: BCRP. Retrieved from <https://www.bcrp.gob.pe/publicaciones/memoria-anual.html>
- Cuaresma JC (2003) Okun's law revisited. *Oxford Bull Econ Stat* 65(4):439–451. <https://doi.org/10.1111/1468-0084.t01-1-00056>
- Dell'Anno R, y Solomon OH (2008) Shadow economy and unemployment rate in USA: is there a structural relationship? An empirical analysis. *App Econ* 40(19):2537–2555. <https://doi.org/10.1080/00036840600970195>
- Dickey DA, y Fuller WA (1981) Likelihood ratio tests for autoregressive time series with a unit root. *Econometrica* 4:1057–1072. <https://doi.org/10.2307/1912517>
- Diebold FX, y Yilmaz K (2012) Better to give than to receive: Predictive directional measurement of volatility spillovers. *Int J Forecasting* 28(1):57–66. <https://doi.org/10.1016/j.ijforecast.2011.02.006>
- Dietsche E (2020) Jobs, skills and the extractive industries: a review and situation analysis. *Miner Econ* 33(3):359–373. <https://doi.org/10.1007/s13563-020-00219-2>
- Evans G (1989) Output and Unemployment Dynamics in the United States: 1950–1985. *J Appl Econ* John Wiley and Sons Ltd 4(3):213–237. <https://doi.org/10.1002/jae.3950040302>
- Feenstra RC, Inklaar R, Timmer MP (2015) The Next Generation of the Penn World Table. *Am Econ Rev* 105(10):3150–3182. Recuperado de <https://www.ggd.net/pwt> <https://doi.org/10.15141/S50TOR>
- Freeman DG (2001) Panel tests of Okun's law for ten industrial countries. *Econ Inq* 39(4):511–523. <https://doi.org/10.1093/ei/39.4.511>
- Garavito C (2003) La ley de Okun en el Perú: 1970–2000. *Economía* 26(51–52):157–196. Recuperado de <http://revistas.pucp.edu.pe/index.php/economia/article/view/731>
- Gordon R (2010) “Okun's Law, Productivity Innovations, and Conundrums in Business Cycle Dating. *Am Econ Rev* 100(2):1–9. Recuperado de <https://pubs.aeaweb.org/doi/pdf/10.1257/aer.100.2.11>
- Grant AP (2002) Time-Varying estimates of the Natural Rate of Unemployment: a revisitation of Okun's Law”. *Quart Rev Econ Finance* 42(spring):95–113. [https://doi.org/10.1016/S1062-9769\(01\)00109-0](https://doi.org/10.1016/S1062-9769(01)00109-0)
- Harris R (2001) Testing for asymmetry in Okun's law: a cross-country comparison. *Econ Bull* 5(2):1–13. Recuperado de <http://eprints.gla.ac.uk/15512/>
- Hodrick RJ, y Prescott EC (1997) Postwar U. S. Business Cycles : An Empirical Investigation. *J Money Credit Bank* 29(1):1–16. <https://doi.org/10.2307/2953682>
- Holmes MJ, y Silverstone B (2006) Okun's law, asymmetries and jobless recoveries in the United States: A Markov-switching approach. *Economics Letters* 92(2):293–299. Recuperado de <https://hdl.handle.net/10289/1879>
- INEI - Instituto Nacional de Estadística e Informática de Perú (2019) Recuperado de <http://www.inei.gob.pe/>
- Jiménez Villavicencio K, y Ochoa Moreno, WS (2017) Ley de Okun. Análisis de la relación entre crecimiento y desempleo para 12 países de América Latina. *Revista Publicando*, 4(13 (1), 1–15. Retrieved from <https://revistapublicando.org/revista/index.php/crv/article/view/773>
- Kahraman E, y Akay O (2023) Comparison of exponential smoothing methods in forecasting global prices of main metals. *Mineral Economics*, 36(3), 427–435. <https://doi.org/10.1007/s13563-022-00354-y>
- Karfakis C, Katrakilidis C, y Tsanana E (2014) Does output predict unemployment? A look at Okun's law in Greece. *International Labour Review* 153(3):421–433. <https://doi.org/10.1111/j.1564-913X.2014.00018.x>
- Knoester A (1986) Okun 's Law Revisited. *Weltwirtschaftliches Archiv* 122(4):657–666 (Recuperado de <https://www.jstor.org/stable/40439470>)
- Lee J (2000) The Robustness of Okun's Law: Evidence from OECD Countries. *J Macroecon* 22(2):331–356. [https://doi.org/10.1016/S0164-0704\(00\)00135-X](https://doi.org/10.1016/S0164-0704(00)00135-X)
- Loayza NV, Raddatz C (2007) The structural determinants of external vulnerability. *World Bank Econ Rev* 21(3):359–387. <https://doi.org/10.1093/wber/lhm018>
- Rodríguez López, Patricia, Peredo y Rodríguez, Felipe de Jesús (2007). Estimación de la Ley de Okun para la economía mexicana. Análisis Económico, XXII(51), 59–79. [fecha de Consulta 11 de Febrero de 2021]. ISSN: 0185–3937. Disponible en: <https://www.redalyc.org/articulo.oa?id=413/41311486004>
- Melguizo C (2017) An analysis of Okun's law for the Spanish provinces. *Rev Regional Res* 37:59–90. <https://doi.org/10.1007/s10037-016-0110-7>
- Miller S (2003) Métodos alternativos para la estimación del PBI potencial: Una aplicación para el caso de Perú. *Estudios Económicos*, 10, 1–38. Recuperado de <https://www.bcrp.gob.pe/docs/Publicaciones/Revista-Estudios-Economicos/10/Estudios-Economicos-10-3.pdf>
- Moosa I (1997) A Cross-Country Comparison of Okun's Coefficient. *J Comp Econ* 24(3):335–356. <https://doi.org/10.1006/jcec.1997.1433>
- Okun AM (1962) Potential GNP: Its measurement and significance. Reprinted from the 1962 proceeding of the business and Economic Statistic Section of the American Statistical Association. Coles Foundation Paper 190, Yale University, 98–104. Recuperado de <https://milesorak.com/okun-potential-gnp-its-measurement-and-significance-p0190/>
- Molero Oliva, Leobaldo Enrique (2012) Estimación de la Ley de Okun para la economía venezolana. Período 1999–2009. *Revista de Ciencias Sociales (Ve)*, XVIII(2), 311–324. ISSN:1315–9518. Recuperado de <https://www.redalyc.org/articulo.oa?id=280/28023310011>
- Owyang MT, Sekhposyan T (2012) Okun's law over the business cycle: was the great recession all that different? Federal Reserve Bank of St. Louis. <https://doi.org/10.20955/r.94.399-418>
- Phillips PC, y Perron P (1988) Testing for a unit root in time series regression. *Biometrika* 75(2):335–346. <https://doi.org/10.1093/biomet/75.2.335>
- Prachowny Martin FJ (1993) “Okun's Law: Theoretical Foundations and Revised Estimates. *The Review of Economics and Statistics* MIT Press 75(2): 331–336. Recuperado de <https://www.jstor.org/stable/2109440>
- Raddatz C (2008) External shocks and macroeconomic volatility in Latin America. World Bank Policy Research Working Paper, 4345. Retrieved from: <https://www.semanticscholar.org/paper/External-Shocks-and-Macroeconomic-Volatility-in-Raddatz/b737b06db1aa693aae903d851fe0d8115c7402e5>
- Renner S, Wellmer FW (2020) Volatility drivers on the metal market and exposure of producing countries. *Miner Econ* 33(3):311–340. <https://doi.org/10.1007/s13563-019-00200-8>

- Sánchez-Fung JR (2015) Producto, desempleo y la Ley de Okun en la República Dominicana. *Ciencia y Sociedad* 40(3):613–637 (Recuperado de <https://doi.org/10.22206/cys.2015.v40i3.pp613-637>)
- Sandrine Sher Verick, y Fares Al-Hussami (2012) Diverging trends in unemployment in the United States and Europe: Evidence from Okun's law and the global financial crisis. *Employment Working Papers*, ILO. Recuperado de: https://www.ilo.org/employment/Whatwedo/Publications/working-papers/WCMS_170782/lang-en/index.htm
- Silvapulle P, Moosa IA, y Silvapulle MJ (2004) Asymmetry in Okun's law. *Canadian J Econ/Revue canadienne d'économie* 37(2):353–374. <https://doi.org/10.1111/j.0008-4085.2004.00006.x>
- Smith G (1975) Okun's law revisited. *Quarterly Rev Econ Bus* 15:37–54 (Recuperado de <https://cowles.yale.edu/sites/default/files/files/pub/d03/d0380.pdf>)
- Sobrinho CR (2011) Current account, productivity and terms of trade shocks in Norway. *Appl Econ Lett* 18(18):1745–1750. <https://doi.org/10.1080/13504851.2011.562154>
- Sögner L, y Stiassny A (2002) An analysis on the structural stability of Okun's law—a cross-country study. *Applied Economics* 34(14):1775–1787. Recuperado de: <https://www.tandfonline.com/doi/abs/10.1080/00036840210124180>
- Sohrabi P, Jodeiri Shokri B, Dehghani H (2023) Predicting coal price using time series methods and combination of radial basis function (RBF) neural network with time series. *Miner Econ* 36(2):207–216. <https://doi.org/10.1007/s13563-021-00286-z>
- Vásquez D (2018) Asimetrías de la Ley de Okun para América Latina. 1980–2016. Repositorio Institucional: Universidad De La Cuenca, 1–112. Recuperado de <http://dspace.ucuenca.edu.ec/handle/123456789/28903>
- Villaverde J, y Maza A (2009) The robustness of Okun's law in Spain, 1980–2004: Regional evidence. *J Policy Model* 31(2):289–297. <https://doi.org/10.1016/j.jpolmod.2008.09.003>
- Weber C (1995) Cyclical output, cyclical unemployment, and Okun's coefficient: A new approach *10* (4): 433–445. <https://doi.org/10.1002/jae.3950100407>
- Zhong M, He R, Chen J, Huang J (2019) Time-varying effects of international nonferrous metal price shocks on China's industrial economy. *Physica A* 528:121299. <https://doi.org/10.1016/j.physa.2019.121299>

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